

Summer School Lecture Notes: Semidefinite Programming in Quantum Information

Lecture 1: Notation, Basic Objects, and Duality for SDPs

Reading (inspiration): These notes follow standard conventions in quantum information and semidefinite programming; they are inspired by lecture notes of John Watrous.

1 Scope and goals of Lecture 1

This lecture sets notation and recalls the basic linear-algebraic objects of quantum information (density matrices, positive semidefinite matrices, Hermitian matrices, and completely positive maps). We then explain why convex optimization is pervasive in quantum information, introduce semidefinite programs (SDPs) as a special case of conic programming, and develop the Lagrangian viewpoint that leads to SDP duality.

2 Notation and linear algebra basics

2.1 Vector spaces and operators

Throughout, $\mathcal{X}$ and $\mathcal{Y}$ denote finite-dimensional complex Hilbert spaces. We write $L(\mathcal{X})$ for the space of linear operators acting on $\mathcal{X}$.

Definition 2.1 (Hermitian and positive semidefinite operators). An operator $X \in L(\mathcal{X})$ is *Hermitian* if $X = X^*$, and we write $X \in \text{Herm}(\mathcal{X})$. It is *positive semidefinite* (PSD) if $X \succeq 0$, meaning $\langle v, Xv \rangle \geq 0$ for all $v \in \mathcal{X}$; we write $X \in \text{Pos}(\mathcal{X})$.

Remark 2.2. If $X \in \text{Herm}(\mathcal{X})$ then it has a real spectrum and an orthonormal eigenbasis. If $X \succeq 0$ then all eigenvalues are nonnegative.

2.2 Trace, Hilbert–Schmidt inner product, and norms

For $X, Y \in L(\mathcal{X})$ we use the Hilbert–Schmidt inner product

$$\langle X, Y \rangle = \text{Tr}(X^*Y).$$

When X, Y are Hermitian, this reduces to $\langle X, Y \rangle = \text{Tr}(XY)$.

Definition 2.3 (Density operators). A *density operator* (or *quantum state*) on $\mathcal{X}$ is an operator $\rho \in D(\mathcal{X})$ defined by

$$D(\mathcal{X}) = \{ \rho \in \text{Pos}(\mathcal{X}) : \text{Tr}(\rho) = 1 \}.$$

Example 2.4 (Pure states). If $|\psi\rangle \in \mathcal{X}$ is a unit vector, then $\rho = |\psi\rangle\langle\psi|$ is a density operator. These are the *pure* states.

2.3 Convexity

Definition 2.5 (Convex sets and convex functions). A set C in a vector space is *convex* if for all $x, y \in C$ and $t \in [0, 1]$, the point $tx + (1 - t)y$ is in C . A function f is *convex* if

$$f(tx + (1 - t)y) \leq tf(x) + (1 - t)f(y) \quad \text{for all } x, y \text{ and } t \in [0, 1].$$

Remark 2.6. The set of density operators $D(\mathcal{X})$ is convex. In quantum information, many tasks reduce to optimizing a linear functional over a convex set of states or measurements, which is precisely the domain of convex optimization.

3 Linear maps, complete positivity, and adjoints

3.1 Linear maps and the adjoint (dual) map

Let $\Phi : L(\mathcal{X}) \rightarrow L(\mathcal{Y})$ be a linear map. Its *adjoint* (also called the *dual map* with respect to the Hilbert–Schmidt inner product) is the unique linear map $\Phi^* : L(\mathcal{Y}) \rightarrow L(\mathcal{X})$ satisfying

$$\langle Y, \Phi(X) \rangle = \langle \Phi^*(Y), X \rangle \quad \text{for all } X \in L(\mathcal{X}) \text{ and } Y \in L(\mathcal{Y}).$$

Remark 3.1. In particular, if $\varphi : L(\mathcal{X}) \rightarrow \mathbb{R}$ is a real-valued linear functional on Hermitian matrices, then there exists a unique $H \in \text{Herm}(\mathcal{X})$ such that $\varphi(X) = \text{Tr}(HX)$ for all $X \in \text{Herm}(\mathcal{X})$. Indeed, the real vector space $\text{Herm}(\mathcal{X})$ becomes a finite-dimensional Hilbert space when equipped with the Hilbert–Schmidt inner product

$$\langle A, B \rangle = \text{Tr}(AB) \quad (A, B \in \text{Herm}(\mathcal{X})).$$

By the Riesz representation theorem for Hilbert spaces, every *continuous* linear functional φ on a Hilbert space has the form

$$\varphi(X) = \langle H, X \rangle$$

for a unique H in the same space. In finite dimensions, every linear map is automatically continuous, so this applies to any linear functional on $\text{Herm}(\mathcal{X})$. Finally, $\langle H, X \rangle = \text{Tr}(HX)$ for Hermitian H, X , giving the claimed representation.

3.2 Completely positive maps

Definition 3.2 (Positive and completely positive maps). A linear map $\Phi : L(\mathcal{X}) \rightarrow L(\mathcal{Y})$ is *positive* if $X \succeq 0 \implies \Phi(X) \succeq 0$. It is *completely positive* (CP) if $\Phi \otimes \text{id}_{\mathcal{Z}}$ is positive for every auxiliary space $\mathcal{Z}$.

Remark 3.3. Quantum channels are completely positive and trace-preserving (CPTP) maps.

4 Why convex optimization (and SDPs) matter in quantum information

A recurring pattern in quantum information is: the feasible objects (states, measurements, channels) form convex sets, and many operational quantities are linear or convex in the decision variables.

This makes convex optimization a natural language for:

- optimal hypothesis testing / state discrimination,
- entanglement detection and quantification,
- channel capacities and coding converses (via relaxations),
- bounding norms (trace norm, operator norm) and eigenvalues.

5 Semidefinite programs (SDPs)

5.1 What an SDP is (geometric/algebraic viewpoint)

At a high level, an SDP is a convex optimization problem in which the variable is a Hermitian matrix X , the objective is *linear* in X , the equality/inequality constraints are *linear* in X , and the “nonlinear” part of the feasible region comes only from requiring that X be positive semidefinite.

Concretely, the basic building blocks are:

- **Linear objective:** maximize (or minimize) $\langle C, X \rangle = \text{Tr}(CX)$ for a fixed Hermitian matrix C ;
- **Linear constraints:** enforce equations/inequalities of the form $\langle A_i, X \rangle = b_i$ (or $\leq b_i$), which are just linear constraints on the entries of X ;
- **PSD constraint:** require $X \succeq 0$, i.e. X has nonnegative eigenvalues.

It is often helpful to think of an SDP as “linear programming, but where the nonnegativity constraint $x \geq 0$ is replaced by the matrix constraint $X \succeq 0$.” In quantum information, this matches the fact that physical states and measurement operators are required to be PSD.

5.2 Historical context

The intellectual ingredients behind semidefinite programming are older than the name itself: convexity, Lagrange multipliers, and duality ideas were developed throughout the first half of the 20th century, and the spectral properties of PSD matrices were (of course) already fundamental in linear algebra. What changed over time was the recognition that *matrix inequalities* such as $X \succeq 0$ can play the same role in optimization as scalar inequalities, and that doing so unlocks a large class of applications.

From linear programming to convex optimization. Linear programming (LP) became a central algorithmic tool in the 1940s–1950s, and it established an enduring template: optimize a linear objective subject to linear constraints and a simple convex feasibility condition (nonnegativity of variables). Over subsequent decades, researchers realized that many problems are still tractable if one replaces the nonnegativity constraint by other structured convex constraints. In this sense, SDPs are part of the broader “convex optimization revolution,” where one aims to keep the objective and constraints convex while enlarging the modeling language.

The algorithmic turning point: interior-point methods. While SDPs can be written down abstractly, they became practically important when polynomial-time algorithms with good numerical behavior were developed. The decisive step was the extension of *interior-point methods* (originally developed for LP) to general convex programs with self-concordant barrier functions, and in particular to SDPs. This happened in the late 1980s and 1990s and led to robust solvers that could handle moderate-to-large instances reliably. From the perspective of these notes, we will treat this as a black box: SDPs are not only convex, but they also come with an algorithmic toolkit that is both theoretically well-understood and practically effective.

Why SDPs spread: relaxations and certificates. Another reason SDPs gained influence is that they provide a systematic way to produce *convex relaxations* of hard (often discrete or non-convex) problems, together with dual solutions that act as *certificates* (upper/lower bounds) of optimality for the relaxation. In combinatorial optimization, for example, many famous approximation algorithms can be viewed as solving an SDP relaxation and then rounding. In control theory and signal processing, SDPs appear through linear matrix inequalities (LMIs) that express stability, robustness, and energy constraints.

Connection to quantum information. In quantum information, positivity constraints are not a modeling trick but a physical requirement: states satisfy $\rho \succeq 0$ and measurements satisfy $M_a \succeq 0$. As a result, many operational tasks—optimal discrimination, hypothesis testing, bounding distinguishability measures, entanglement witnesses/relaxations, and channel optimizations—are naturally expressed as SDPs. This is one reason SDP duality is so useful in the field: the dual problem often has a direct operational interpretation (e.g. a witness, a bound, or a “pretty” inequality) and can be used to prove impossibility results.

Sums of squares (SOS) relaxations and SDP hierarchies (e.g. NPA). Beyond “one-shot” SDPs, an important theme in modern optimization is to approximate difficult nonconvex polynomial problems by a *sequence* of SDPs of increasing size and strength. In the polynomial-optimization world, the key idea is a *sums-of-squares* (SOS) certificate: proving a polynomial is nonnegative by writing it as a sum of squares, which can be enforced via a Gram-matrix PSD constraint and hence reduced to an SDP. In quantum nonlocality and device-independent quantum information, a closely related role is played by SDP hierarchies such as the Navascués–Pironio–Acín (NPA) hierarchy: one considers moment matrices built from words in measurement operators and imposes positivity/consistency constraints; truncating at higher word length yields systematically tighter SDP outer approximations to the set of quantum correlations. These hierarchies illustrate a broader point: even when the exact set of feasible quantum behaviors is complicated, SDP relaxations often provide convergent, computable bounds.

5.3 Standard primal/dual SDP forms

We are now ready to finally define what an SDP is. One convenient *standard form* (many equivalent conventions exist) is:

Primal (P).

$$\begin{aligned} & \text{maximize: } \langle C, X \rangle \\ & \text{subject to: } \langle A_i, X \rangle = b_i \quad (i = 1, \dots, m) \\ & \quad X \succeq 0, \end{aligned}$$

where $X \in \text{Herm}(\mathcal{X})$ is the variable, $C, A_1, \dots, A_m \in \text{Herm}(\mathcal{X})$ and $b \in \mathbb{R}^m$ are given.

Another important optimization problem associated to an SDP is its dual. We will understand shortly where it comes from, let us just define it now:

Dual (D).

$$\begin{aligned} & \text{minimize: } b^\top y \\ & \text{subject to: } \sum_{i=1}^m y_i A_i \succeq C \\ & \quad y \in \mathbb{R}^m. \end{aligned}$$

5.4 Deriving the dual in standard form

We now briefly justify the dual (D) from the primal (P) using the same Lagrangian reasoning that we will apply repeatedly later.

Form the Lagrangian by introducing multipliers $y \in \mathbb{R}^m$ for the equality constraints and a slack matrix $S \succeq 0$ for the PSD constraint:

$$\mathcal{L}(X, y, S) = \langle C, X \rangle + \sum_{i=1}^m y_i (b_i - \langle A_i, X \rangle) - \langle S, X \rangle = b^\top y + \left\langle C - \sum_{i=1}^m y_i A_i - S, X \right\rangle.$$

If $C - \sum_i y_i A_i - S \neq 0$, then the supremum of $\mathcal{L}(X, y, S)$ over $X \in \text{Herm}(\mathcal{X})$ is $+\infty$ (one can scale X in the direction of the mismatch). Hence, for the dual function to be finite we must have

$$C - \sum_{i=1}^m y_i A_i - S = 0 \quad \Longleftrightarrow \quad \sum_{i=1}^m y_i A_i \succeq C,$$

where the equivalence uses $S \succeq 0$. Under this condition, $\sup_X \mathcal{L}(X, y, S) = b^\top y$, and minimizing over (y, S) gives exactly the dual SDP (D).

Remark 5.1 (Polynomial-time solvability (informal)). SDPs can be solved in time polynomial in the input size to any prescribed accuracy under mild regularity conditions. We will treat algorithms as a black box for now and focus on modeling and duality. We then discuss how to solve them in practice in Lecture 3.

Before we discuss duality theory, let us discuss some simple examples of SDPs.

5.5 Example: largest eigenvalue as an SDP

Let $M \in \text{Herm}(\mathcal{X})$. Recall the variational characterization

$$\lambda_{\max}(M) = \max\{ \langle v, Mv \rangle : \|v\|_2 = 1 \} = \max\{ \text{Tr}(M\rho) : \rho \in \mathcal{D}(\mathcal{X}) \}.$$

The second expression is an SDP because $\rho \succeq 0$ and $\text{Tr}(\rho) = 1$ are affine/semidefinite constraints and the objective is linear:

$$\begin{aligned} & \text{maximize: } \text{Tr}(M\rho) \\ & \text{subject to: } \text{Tr}(\rho) = 1 \\ & \quad \rho \succeq 0. \end{aligned}$$

Thus, we see that finding the largest eigenvalue of a matrix can be cast as an SDP.

5.6 Example: minimum-error quantum state discrimination

Let $\{\rho_a\}_{a=1}^k \subset \mathcal{D}(\mathcal{X})$ be candidate states prepared with prior probabilities $\{p_a\}_{a=1}^k$. A measurement is a POVM $\{M_a\}_{a=1}^k$ with $M_a \succeq 0$ and $\sum_a M_a = \mathbb{1}$. The success probability is

$$p_{\text{succ}} = \sum_{a=1}^k p_a \text{Tr}(M_a \rho_a).$$

Maximizing p_{succ} over POVMs is an SDP:

$$\begin{aligned} & \text{maximize: } \sum_{a=1}^k p_a \text{Tr}(M_a \rho_a) \\ & \text{subject to: } \sum_{a=1}^k M_a = \mathbb{1} \\ & \quad M_a \succeq 0 \quad (a = 1, \dots, k). \end{aligned}$$

Operational interpretation. operationally, this SDP is modelling the task of designing the best measurement for guessing which state was prepared. Indeed, imagine an experimenter (Alice) who samples a label $a \in \{1, \dots, k\}$ according to the prior distribution $\{p_a\}$, prepares the corresponding quantum state ρ_a , and sends it to a second experimenter (Bob). Bob chooses a POVM $\{M_a\}$ and outputs the outcome label as his guess $\hat{a}$.

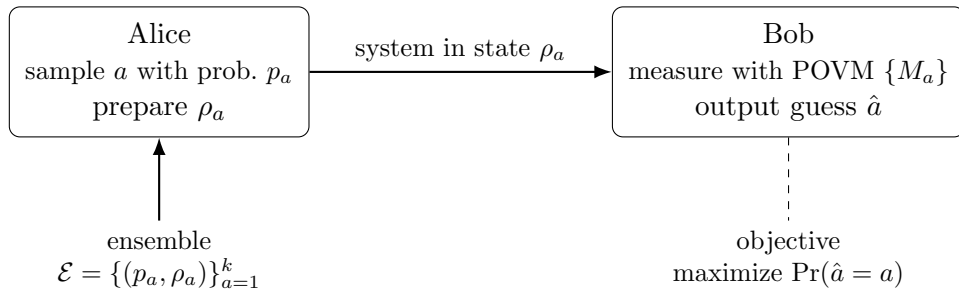


Figure 1: State discrimination: Alice samples a label a and prepares ρ_a from an ensemble $\mathcal{E}$; Bob applies a POVM $\{M_a\}$ and outputs a guess $\hat{a}$, aiming to maximize the success probability $\Pr(\hat{a} = a)$.

The Born rule gives

$$\Pr(\hat{a} = a \mid a) = \text{Tr}(M_a \rho_a),$$

so by averaging over the prior we obtain the success probability

$$\Pr(\hat{a} = a) = \sum_{a=1}^k p_a \operatorname{Tr}(M_a \rho_a),$$

which is exactly the objective function. The constraints $M_a \succeq 0$ and $\sum_a M_a = \mathbb{1}$ are precisely the physical constraints that $\{M_a\}$ define a valid measurement with k outcomes. Therefore, the optimal value of the SDP is the *Helström optimal* minimum-error discrimination probability for the ensemble $\{(p_a, \rho_a)\}$, and any optimal solution $\{M_a^*\}$ specifies an optimal measurement strategy.

Remark 5.2. Later, when we derive the dual SDP, its dual variable can be interpreted as an operator that upper bounds each weighted state $p_a \rho_a$. This yields an alternative characterization of the optimum and is often the cleanest way to prove analytic bounds on discrimination probability.

6 Lagrange multipliers and duality for SDPs

6.1 Lagrangian viewpoint

For the primal SDP

$$\max\{\langle C, X \rangle : \langle A_i, X \rangle = b_i \ (i = 1, \dots, m), \ X \succeq 0\},$$

introduce Lagrange multipliers $y \in \mathbb{R}^m$ for the equality constraints and a PSD “slack” matrix $S \succeq 0$ for the cone constraint. Consider the Lagrangian

$$\mathcal{L}(X, y, S) = \langle C, X \rangle + \sum_{i=1}^m y_i (b_i - \langle A_i, X \rangle) - \langle S, X \rangle.$$

Rearranging terms gives

$$\mathcal{L}(X, y, S) = b^\top y + \left\langle C - \sum_i y_i A_i - S, X \right\rangle.$$

Taking the supremum over X forces $C - \sum_i y_i A_i - S = 0$, i.e.,

$$\sum_{i=1}^m y_i A_i \succeq C,$$

and then the Lagrangian supremum equals $b^\top y$. This yields the dual program (D) above.

6.2 Duals of the examples

We now derive the dual SDPs for the two examples above. This serves two purposes: (i) it provides additional practice in forming Lagrangians, and (ii) in quantum information the *dual* often has the cleanest operational meaning (as a witness or a bound).

6.2.1 Dual of the largest-eigenvalue SDP

Consider the primal

$$\begin{aligned} & \text{maximize: } \text{Tr}(M\rho) \\ & \text{subject to: } \text{Tr}(\rho) = 1 \\ & \quad \rho \succeq 0. \end{aligned}$$

Introduce a scalar Lagrange multiplier $\lambda \in \mathbb{R}$ for the constraint $\text{Tr}(\rho) = 1$ and a PSD slack matrix $S \succeq 0$ for $\rho \succeq 0$. The Lagrangian is

$$\mathcal{L}(\rho, \lambda, S) = \text{Tr}(M\rho) + \lambda(1 - \text{Tr}(\rho)) - \text{Tr}(S\rho) = \lambda + \text{Tr}((M - \lambda\mathbb{1} - S)\rho).$$

Taking the supremum over ρ forces $M - \lambda\mathbb{1} - S = 0$, i.e. $\lambda\mathbb{1} \succeq M$, and then the value becomes λ . Thus the dual is

$$\begin{aligned} & \text{minimize: } \lambda \\ & \text{subject to: } \lambda\mathbb{1} \succeq M, \\ & \quad \lambda \in \mathbb{R}. \end{aligned}$$

This recovers the familiar characterization $\lambda_{\max}(M) = \min\{\lambda : \lambda\mathbb{1} \succeq M\}$.

6.2.2 Dual of the minimum-error discrimination SDP

The primal is

$$\begin{aligned} & \text{maximize: } \sum_{a=1}^k p_a \text{Tr}(M_a \rho_a) \\ & \text{subject to: } \sum_{a=1}^k M_a = \mathbb{1}, \\ & \quad M_a \succeq 0 \quad (a = 1, \dots, k). \end{aligned}$$

Introduce a Hermitian Lagrange multiplier $H \in \text{Herm}(\mathcal{X})$ for the operator constraint $\sum_a M_a = \mathbb{1}$, and slack matrices $S_a \succeq 0$ for $M_a \succeq 0$. The Lagrangian is

$$\mathcal{L}(\{M_a\}, H, \{S_a\}) = \sum_{a=1}^k p_a \text{Tr}(M_a \rho_a) + \text{Tr}\left(H\left(\mathbb{1} - \sum_{a=1}^k M_a\right)\right) - \sum_{a=1}^k \text{Tr}(S_a M_a).$$

Grouping the terms involving M_a yields

$$\mathcal{L}(\{M_a\}, H, \{S_a\}) = \text{Tr}(H) + \sum_{a=1}^k \text{Tr}((p_a \rho_a - H - S_a)M_a).$$

Taking the supremum over the (unconstrained) Hermitian matrices M_a forces

$$p_a \rho_a - H - S_a = 0 \quad \text{for each } a,$$

which is equivalent to the inequalities $H \succeq p_a \rho_a$ for all a (because each $S_a \succeq 0$). The dual therefore becomes

$$\begin{aligned} & \text{minimize: } \text{Tr}(H) \\ & \text{subject to: } H \succeq p_a \rho_a \quad (a = 1, \dots, k), \\ & \quad H \in \text{Herm}(\mathcal{X}). \end{aligned}$$

Remark 6.1 (Interpretation). The constraint $H \succeq p_a \rho_a$ says that H upper bounds each weighted state $p_a \rho_a$ in the semidefinite order, and the objective $\text{Tr}(H)$ is the smallest trace among all such upper bounds. Weak duality immediately gives an upper bound on the achievable discrimination probability, and when strong duality holds this bound is tight.

6.3 Weak and strong duality

Proposition 6.2 (Weak duality). *For any primal feasible X and dual feasible y , one has $\langle C, X \rangle \leq b^\top y$.*

Proof. Let X be primal feasible, so $X \succeq 0$ and $\langle A_i, X \rangle = b_i$ for all i . Let y be dual feasible, so $\sum_{i=1}^m y_i A_i \succeq C$, i.e. $S := \sum_{i=1}^m y_i A_i - C \succeq 0$. Taking the Hilbert–Schmidt inner product with $X \succeq 0$ gives

$$0 \leq \langle S, X \rangle = \left\langle \sum_{i=1}^m y_i A_i - C, X \right\rangle = \sum_{i=1}^m y_i \langle A_i, X \rangle - \langle C, X \rangle = b^\top y - \langle C, X \rangle.$$

Rearranging yields $\langle C, X \rangle \leq b^\top y$. □

6.4 Slater’s theorem (strong duality)

Weak duality always holds, but strong duality requires additional regularity. The most common and useful sufficient condition is *strict feasibility*, often called *Slater’s condition*.

Theorem 6.3 (Slater’s condition for SDPs (statement)). *Consider the primal/dual pair in standard form:*

$$\begin{array}{ll} (P) & \text{maximize: } \langle C, X \rangle \\ & \text{subject to: } \langle A_i, X \rangle = b_i \quad (i = 1, \dots, m), \\ & \quad \quad \quad X \succeq 0, \\ (D) & \text{minimize: } b^\top y \\ & \text{subject to: } \sum_{i=1}^m y_i A_i \succeq C, \\ & \quad \quad \quad y \in \mathbb{R}^m. \end{array}$$

Assume the primal is feasible and there exists a strictly feasible point $X_0 \succ 0$ such that $\langle A_i, X_0 \rangle = b_i$ for all i (primal Slater condition). Then:

- (i) *the dual optimum is attained (there exists an optimal y^*), and*
- (ii) *the optimal values coincide: $\max(P) = \min(D)$.*

Similarly, if the dual is feasible and there exists y_0 such that $\sum_i (y_0)_i A_i \succ C$ (dual Slater condition), then the primal optimum is attained and $\max(P) = \min(D)$.

Remark 6.4. This proof is slightly more involved. IT is ok if you do not fully understand it. What is important is that you keep Slater’s condition in mind!

Proof. We prove the first implication: if the primal is feasible and there exists $X_0 \succ 0$ with $\langle A_i, X_0 \rangle = b_i$ for all i , then there is no duality gap and the dual optimum is attained. The second implication (dual strict feasibility) is analogous.

Define the linear map $\mathcal{A} : \text{Herm}(\mathcal{X}) \rightarrow \mathbb{R}^m$ by

$$\mathcal{A}(X) = (\langle A_1, X \rangle, \dots, \langle A_m, X \rangle),$$

and its adjoint $\mathcal{A}^* : \mathbb{R}^m \rightarrow \text{Herm}(\mathcal{X})$ by $\mathcal{A}^*(y) = \sum_{i=1}^m y_i A_i$. Write the primal optimum as

$$\alpha := \sup\{\langle C, X \rangle : \mathcal{A}(X) = b, X \succeq 0\}.$$

Step 1: set up a separation argument. Consider the convex set

$$\mathcal{K} := \{(\mathcal{A}(X), \langle C, X \rangle) : X \succeq 0\} \subset \mathbb{R}^{m+1}.$$

By definition of α , for any $\varepsilon > 0$ the point $(b, \alpha + \varepsilon)$ does not belong to $\mathcal{K}$. By the separating hyperplane theorem, there exist $(y, \mu) \in \mathbb{R}^m \times \mathbb{R}$, not both zero, and $\beta \in \mathbb{R}$ such that

$$y^\top u + \mu v \leq \beta \quad \text{for all } (u, v) \in \mathcal{K}, \quad \text{and} \quad y^\top b + \mu(\alpha + \varepsilon) > \beta.$$

Unpacking $(u, v) = (\mathcal{A}(X), \langle C, X \rangle)$ yields, for all $X \succeq 0$,

$$y^\top \mathcal{A}(X) + \mu \langle C, X \rangle \leq \beta.$$

Step 2: show that $\mu < 0$. We claim that $\mu < 0$. If $\mu > 0$, then scaling $X \mapsto tX$ for $t \rightarrow \infty$ would force

$$y^\top \mathcal{A}(X) + \mu \langle C, X \rangle \leq 0 \quad \text{for all } X \succeq 0,$$

otherwise the left-hand side would grow without bound and violate the inequality. In particular, evaluating at $X = X_0 \succ 0$ gives $y^\top b + \mu \langle C, X_0 \rangle \leq 0$, hence $y^\top b < 0$. But plugging $X = 0$ into the inequality gives $0 \leq \beta$, and plugging $X = X_0$ gives $y^\top b + \mu \langle C, X_0 \rangle \leq \beta$, implying $y^\top b < \beta$. This contradicts the strict separation $y^\top b + \mu(\alpha + \varepsilon) > \beta$ for sufficiently small $\varepsilon > 0$ (because $\mu > 0$ would make the left-hand side arbitrarily close to $y^\top b$ as $\varepsilon \downarrow 0$). Thus $\mu \leq 0$.

If $\mu = 0$, the inequality becomes $y^\top \mathcal{A}(X) \leq \beta$ for all $X \succeq 0$. Scaling $X \mapsto tX$ forces $y^\top \mathcal{A}(X) \leq 0$ for all $X \succeq 0$, hence $y^\top b = y^\top \mathcal{A}(X_0) \leq 0$. But then strict separation would require $y^\top b > \beta$, contradicting $y^\top b \leq \beta$ (from the inequality with $X = X_0$). Therefore $\mu < 0$.

Define $\tilde{y} := -y/\mu \in \mathbb{R}^m$ and $\gamma := -\beta/\mu \in \mathbb{R}$. Dividing by $-\mu > 0$ gives: for all $X \succeq 0$,

$$\langle C, X \rangle \leq \tilde{y}^\top \mathcal{A}(X) + \gamma.$$

Taking $X = 0$ shows $\gamma \geq 0$.

Step 3: dual feasibility. Rewriting the inequality as

$$\langle \mathcal{A}^*(\tilde{y}) - C, X \rangle \geq -\gamma \quad \text{for all } X \succeq 0,$$

and scaling $X \mapsto tX$ for $t \rightarrow \infty$ forces $\langle \mathcal{A}^*(\tilde{y}) - C, X \rangle \geq 0$ for all $X \succeq 0$. This is equivalent to $\mathcal{A}^*(\tilde{y}) - C \succeq 0$, i.e. $\sum_i \tilde{y}_i A_i \succeq C$. Hence $\tilde{y}$ is dual feasible.

Step 4: zero duality gap and attainment. Divide the strict separation by $-\mu > 0$ to obtain

$$\alpha + \varepsilon < \tilde{y}^\top b + \gamma.$$

Letting $\varepsilon \downarrow 0$ yields $\alpha \leq \tilde{y}^\top b + \gamma$. On the other hand, weak duality implies $\alpha \leq b^\top y$ for every dual feasible y , hence in particular $\alpha \leq \tilde{y}^\top b$. Combining gives $\gamma = 0$ and therefore $\alpha \leq \tilde{y}^\top b$ with $\tilde{y}$ dual feasible. Taking the infimum over all dual feasible points yields $\alpha \leq \inf(\mathcal{D})$; together with weak duality ($\alpha \geq \inf(\mathcal{D})$) this implies $\alpha = \inf(\mathcal{D})$ (zero duality gap).

Finally, dual attainment follows because under the primal Slater condition the dual feasible set has bounded level sets: the set

$$\{y : \mathcal{A}^*(y) \succeq C, \ b^\top y \leq \alpha + 1\}$$

is nonempty (it contains $\tilde{y}$), closed, and bounded, hence compact. The continuous function $y \mapsto b^\top y$ therefore attains its minimum on this set, giving an optimal dual solution. $\square$

6.5 Example: failure of Slater's condition and a duality gap

The following example (inspired by standard SDP pathology examples) shows that if strict feasibility fails, strong duality can fail in the most extreme way: the primal has a finite optimum while the dual is infeasible.

Example 6.5 (A duality-gap SDP in dimension 2). Consider the primal SDP

$$\begin{aligned} &\text{minimize: } 2X_{12} \\ &\text{subject to: } X_{11} = 0, \\ &\quad X \succeq 0, \\ &\quad X \in S^2. \end{aligned}$$

Equivalently, in standard form (minimization version) take

$$C = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \mathcal{A}(X) = X_{11} = \langle \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, X \rangle, \quad b = 0.$$

The adjoint map is therefore

$$\mathcal{A}^*(y) = y \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}.$$

Primal value. Any $X \succeq 0$ with $X_{11} = 0$ must have $X_{12} = 0$ (e.g. because every principal minor must be nonnegative, so $\det \begin{pmatrix} 0 & X_{12} \\ X_{12} & X_{22} \end{pmatrix} = -X_{12}^2 \geq 0$ implies $X_{12} = 0$). Hence the objective value is always $2X_{12} = 0$, and so the optimal value is 0.

Dual program (written explicitly). The dual can be written as

$$\begin{aligned} &\text{maximize: } 0 \\ &\text{subject to: } \mathcal{A}^*(y) \preceq C. \end{aligned}$$

Equivalently, introducing a slack variable $Z \succeq 0$ so that $C - \mathcal{A}^*(y) = Z$, we obtain

$$\begin{aligned} & \text{maximize: } 0 \\ & \text{subject to: } \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = y \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + Z, \\ & \quad Z \succeq 0. \end{aligned}$$

But for any real y , the matrix

$$Z = \begin{pmatrix} -y & 1 \\ 1 & 0 \end{pmatrix}$$

has determinant -1 , so it cannot be positive semidefinite. Therefore the dual is infeasible and $d^* = -\infty$, yielding a duality gap.

Remark 6.6. This example is included as a warning: without a condition such as Slater's, one cannot freely exchange primal and dual optima. In practice, when modeling quantum information problems, it is usually easy to check strict feasibility (e.g. by taking a full-rank density matrix or adding a tiny multiple of the identity to a feasible point). *But it is important to check! And many papers do not!*

6.6 Complementary slackness

Complementary slackness is one of the main *structural* consequences of strong duality: it explains how optimal primal and dual solutions “touch” the boundary of the PSD constraint. It gives us more relations that the solutions satisfy.

Recall the standard-form primal/dual pair

$$\begin{aligned} \text{(P)} \quad & \text{maximize: } \langle C, X \rangle \\ & \text{subject to: } \mathcal{A}(X) = b, \\ & \quad X \succeq 0, \\ \text{(D)} \quad & \text{minimize: } b^\top y \\ & \text{subject to: } \mathcal{A}^*(y) \succeq C. \end{aligned}$$

Given a dual feasible y , the associated *dual slack* is

$$S := \mathcal{A}^*(y) - C \succeq 0.$$

Theorem 6.7 (Complementary slackness for SDPs). *Assume strong duality holds and let X^* be primal optimal and y^* be dual optimal. Let*

$$S^* := \mathcal{A}^*(y^*) - C \succeq 0.$$

Then

$$\langle S^*, X^* \rangle = \text{Tr}(S^* X^*) = 0.$$

Equivalently, because $S^ \succeq 0$ and $X^* \succeq 0$, one has*

$$S^* X^* = 0 \quad \text{and hence} \quad \text{supp}(X^*) \subseteq \ker(S^*).$$

Proof. By weak duality, for any primal feasible X and dual feasible y we have

$$\langle C, X \rangle \leq b^\top y.$$

Fix primal feasible X^* and dual feasible y^* and write the dual slack $S^* = \mathcal{A}^*(y^*) - C \succeq 0$. Then

$$b^\top y^* - \langle C, X^* \rangle = \sum_{i=1}^m y_i^* \langle A_i, X^* \rangle - \langle C, X^* \rangle = \langle \mathcal{A}^*(y^*) - C, X^* \rangle = \langle S^*, X^* \rangle.$$

If X^* and y^* are optimal and strong duality holds, then $b^\top y^* = \langle C, X^* \rangle$, so the left-hand side is 0 and thus $\langle S^*, X^* \rangle = 0$.

Finally, for PSD operators $P, Q \succeq 0$, the condition $\text{Tr}(PQ) = 0$ implies $PQ = 0$ (e.g. $0 = \text{Tr}(P^{1/2}QP^{1/2})$ with $P^{1/2}QP^{1/2} \succeq 0$ forces $P^{1/2}QP^{1/2} = 0$, hence $QP^{1/2} = 0$ and so $PQ = 0$). Applying this with $P = S^*$ and $Q = X^*$ yields $S^*X^* = 0$ and the support/kernel inclusion. $\square$

Remark 6.8 (How to use it). Complementary slackness is often used in one of two ways:

- (i) *Proving optimality:* exhibit primal/dual feasible X, y such that $\langle S, X \rangle = 0$; then they are simultaneously optimal.
- (ii) *Extracting structure:* it gives more relations the optimal solution must satisfy, which are useful when proving further properties of them.

7 Further reading

In this lecture, we discussed the basic definitions around SDPs and the basics of duality theory. For more on the structure of SDPs, good references are Watrous' lecture notes still for a QIT perspective and Boyd and Vandenberghe's book for a convex optimization perspective.